DO YOU KNOW?

The following units were named after the scientists, as given below.

Volt	Italian scientist Alessandro volta
Ampere	French scientist André-Marie Ampere
Ohm	German physicist George simon ohm.

1.4 OHM'S LAW

German physicist George Simon Ohm found that "voltage, current and resistance" are related to each other and derived a law. This is called Ohm's law. Let us try to understand ohm's law through the example shown in Figure 1.8.

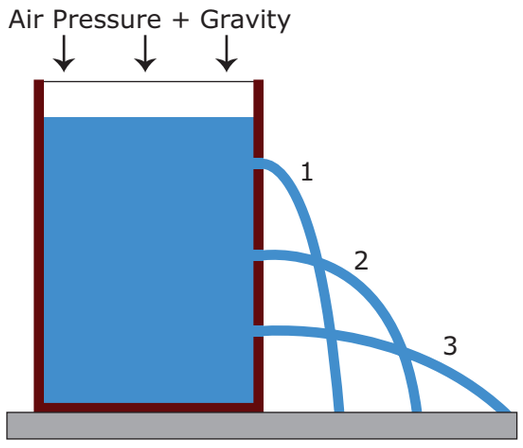


Figure 1.8 **Demonstration of Ohm's law**

Take a plastic bottle and make three equal sizes of holes viz. (i) Top (ii) centre and (iii) bottom and fill up the bottle with water.

What will happen?

There are air pressure and gravitational force always around us. Due to this, water in that bottle is under pressure.

1. From top hole, water flows with a low speed. Because, only air pressure and gravity act on it.
2. Water from the *middle hole* flows with increasing speed than in the top hole, because pressure increases due to water in the top portion of the bottle along with air and gravity.
3. Water from the *bottom hole* flows with high speed than in the first and middle, because pressure increases due to water in the top and middle parts along with air and gravity.

Likewise, in a circuit, if the supply voltage increases, the electric current increases. If the supply voltage decreases, current also decreases (Resistance value should be constant in both the conditions)

$$I \propto V$$

Therefore, the current in a circuit can be represented as,

$$I \alpha \frac{V}{R}$$

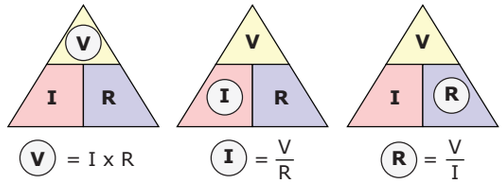
DO YOU KNOW?

Ohm's law states that at constant temperature, the current flows in a circuit is directly proportional to the voltage and inversely proportional to the resistance in the circuit.

$$\text{Current (Ampere)} = \frac{\text{Voltage (Volt)}}{\text{Resistance (ohm}}$$

$$I = \frac{V}{R}$$

The other forms of Ohm's law are given below:



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